Fiscal Year:	FY 2014	**Task Last Updated:**	FY 03/24/2015
PI Name:	Irudayaraj, Joseph Ph.D.		
Project Title:	Defining Epigenetic Programming During Flight Expeditions In Differentiating Embryonic Stem Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	47907-2093	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2015 per A. Chu; original end date was 7/31/2013 (Ed., 7/14/14) NOTE: Extended to 7/31/14 per A. Chu/ARC (Ed., 7/22/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lossie, Amy (Purdue University)		
Grant/Contract No.:	NNX12AN09G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Although scientists can determine the function of every gene, limited methods exist for determining how cells interpret instructions that control gene function. The instructions, (epigenetic marks), encompass chemical modifications added "on top of" DNA that do not cause a genetic mutation. Understanding how diverse patterns of epigenetic marks coordinate the regulation of gene expression is an open question because, although many epigenetic landscapes have been mapped in cell populations, limited strategies have been developed to test and/or understand how epigenetic control occurs in single cells. Epigenetic modifications respond dynamically to environmental factors, and can change within minutes. In space, organisms are exposed to powerful environmental forces (microgravity, cosmic radiation, magnetic fields) that potentially modulate the epigenome, leading to changes in gene expression that affect mammalian development. By elucidating the mechanisms that govern epigenetic programming and gene regulation in single cells, we will provide fundamental knowledge that will broadly impact the epigenetics community, basic biology, and NASA's space biology programs. The goal of our interdisciplinary effort is to develop an integrated approach comprised of Next Generation Sequencing and single cell microscopy platforms to understand how the space environment affects DNA methylation, hydroxymethylation, and histone modifications to regulate gene expression during differentiation using an Embryonic Stem Cell model. Through our interdisciplinary effort, we will create a framework to explain how DNA methylation (5hmeC, 5meC) and histone modifications regulate gene expression not only in response to space flight conditions, but also to environments encountered on Earth. Answers to this question, which must be tested in single cells, have heretofore remained elusive. Upon completion, our work will transform the basic biological understanding of epigenetics, propel the current space biology research program into the realm of epigenetics, and impact the planning of future long-term space flight expeditions at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will constitute one of the first efforts to assess how environmental conditions (i.e. microgravity, radiation) affect epigenetic patterns (i.e. DNA methylation and hydroxymethylation patterns and histone modifications) to coordinate gene expression. We expect these epigenetic events to explain genetic aberrations that cannot be explained by conventional genetic studies. Discovery of site-specific alteration of the epigenome and its role in gene expression will be highly significant not only to the field of epigenetics, but to biology in general. By elucidating epigenetic consequences due to the space flight environment in pluripotent embryonic stem cells (ESCs) and their differentiated counterparts, embryoid bodies, we will begin to discover mechanistically how gravity alters epigenetic modifications and in turn, gene expression in the three basic germlayers (endoderm, ectoderm, and mesoderm) during cellular differentiation. Our ability to complement genomic sequencing with single cell tools will facilitate exploration of any gene at single cell resolution from any organism, a heretofore, elusive feature. With slight modifications, we will have the ability to assess DNA binding factors, reveal functional irregularities based on epigenetic profiles, and screen for drugs that target specific enzymes that add or remove epigenetic modifications. Our methodologies can also be used to follow chromatin through the cell cycle and to determine how DNA damage triggers changes in chromatin. Key applications also include defining real-time chromatin dynamics during cellular growth, differentiation, and early development not only in response to space flight but also diet, nutrition, and alcohol, among other environmental factors. We expect our contributions to impact space biology research in NASA, and the planning of future long-term space flight expeditions.
Task Progress:	HL60 and TK cells were grown in the respective conditions (control and under microgravity condition) and subjected to Next Generation Sequencing (NGS) analysis to determine the methylated regions so that we can enhance our understanding of the gene expression under microgravity conditions. This work fits under the key programmatic goals entitled, “Developmental Biology” (as stated in bullet point 5 on page 7 of NRA NNH11ZZTT002N) and “Cell, Microbial, and Molecular Biology” (as outlined in bullet point 5 of page 10 of the 2010-2010 FSB Plan). Our proposal is also responsive to other goals of the Space Biology; specifically, the proposed experiments and discoveries will “further the basic knowledge relevant to the effect of space flight environment on living systems, maximize the probability of achieving results that can translate into benefits on Earth, and produce knowledge to develop human countermeasures to the detrimental effects encountered in space.” Deliverables of the study that have been accomplished include: 1) Development and optimization of methods for Next Generation Sequencing (MeDIP; hMeDIP) of the two cell lines (ie. HL60 and TK6) 2) Determination of the methylation and hydroxymethylation profiles in the genome and 3) Determination of the key genes that are undergoing change in gene transcription and are significantly methylated/hydroxymethylated by comparative bioinformatics approach on RNA-seq and MeDIP-seq/hMeDIP-seq data- List of Expected Publications 1. Nucleic Acids Research (NAR-00802-2015) “Differentiation-dependent changes in the epigenomic landscape of human myeloid leukemia cells implicate signaling pathways and highlight chromosomal amplifications” Chowdhury et al. [In review] 2. PLOS ONE (PONE-D-15-11894) “A study of alterations in DNA epigenetic modifications (5mC and 5hmC) and gene expression influenced by simulated microgravity in human lymphoblastoid cells” Chowdhury et al. [In Review] 3. Chowdhury, B., Chen, J., Liu, S., Lossie, A., and Irudayaraj, J. 2014. Effect of AZA on hydroxymethylation – exploring a novel effect. Nature Scientific Reports (In Press).
Bibliography Type:	Description: (Last Updated: 06/03/2016)
Articles in Peer-reviewed Journals	Chowdhury B, McGovern A, Cui Y, Choudhury S, Cho I-L, Cooper B, Chevassut T, Lossie A, Irudayaraj J. "The hypomethylating agent Decitabine causes a paradoxical increase in 5-hydroxymethylcytosine in human leukemia cells." Nature Scientific Reports. In press as of March 2015. , Mar-2015